Statement of Purpose – Amazon ML Summer School 2026

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Expected Graduation Year: 2027

My interest in Machine Learning began with a simple question: how can data be transformed into intelligent decisions that solve real-world problems? Over time, this curiosity evolved into a passion for building AI-driven systems that address challenges in healthcare, agriculture, cybersecurity, and enterprise automation.

As a Computer Science student, I have actively explored both the theoretical and practical aspects of Machine Learning through academic projects, research work, and professional certifications. One of my most significant projects involves Motor Imagery EEG Classification, where I worked with brain-signal data to develop deep learning models capable of classifying human motor intentions. This project exposed me to signal processing, feature extraction, neural networks, and the challenges of working with noisy real-world datasets.

I have also applied Machine Learning in cybersecurity through an Intrusion Detection System trained on the CICIDS2017 dataset. This experience strengthened my understanding of data preprocessing, feature engineering, model evaluation, and the importance of building reliable AI systems for critical applications.

Beyond conventional ML applications, I have worked on a privacy-first OCR-LLM ETL pipeline that converts unstructured documents into structured enterprise data using Optical Character Recognition, Retrieval-Augmented Generation, and local language models. Additionally, I have designed AI-enabled solutions for smart farming and intelligent water-dam monitoring systems, combining IoT, predictive analytics, and cloud technologies. These projects taught me that successful AI systems require not only strong models but also robust engineering and domain understanding.

While these experiences have provided valuable practical exposure, I recognize that there are significant gaps in my understanding of advanced Machine Learning concepts. I want to deepen my knowledge of mathematical foundations, optimization techniques, probabilistic modeling, representation learning, and modern deep learning architectures. More importantly, I want to learn how leading researchers approach complex problems and translate cutting-edge research into scalable real-world solutions.

Amazon ML Summer School represents a unique opportunity to bridge this gap. The chance to learn directly from Amazon Scientists, interact with talented peers, and gain exposure to industry-scale ML applications aligns perfectly with my academic and career goals. I believe the program will help me strengthen my fundamentals while providing insights into the practical challenges of deploying Machine Learning systems at scale.

What makes me a strong fit for MLSS is my combination of curiosity, research mindset, and hands-on experience. I enjoy exploring challenging problems, learning new technologies, and applying AI to domains that can create meaningful impact. I am eager to contribute actively to the program, learn from experts and peers, and use the knowledge gained to pursue advanced research and innovation in Machine Learning.

Thank you for considering my application.